sampler. With the aim to reduce this large number, we repeated the SBC simulations for the interaction term with $20,000$ iterations of the MCMC sampler instead of the previous $10,000$ iterations.

We repeated the previous simulations of design 3, but now only for the interaction term. We first investigated model convergence. Across all model fits (200 simulations $\times 2$ models (H0/H1) = 400 fits), we observed 0 divergent transitions. R-hat exceeded the threshold of 1.01 for at least one population-level parameter in 0 cases, suggesting overall good model convergence. However, during the bridge sampling, there was the warning message "logml could not be estimated within maxiter, rerunning with adjusted starting value. Estimate might be more variable than usual" in 34.50% of simulations, suggesting warning messages were somewhat, but not strongly reduced. We again looked at the results separately for simulations with versus without warning messages from the bridge sampler.

The results (see Fig. 4) showed that – for simulations without warning messages from the bridge sampling – it was not clear whether the posterior model probabilities differed from the prior model samples. Indeed, the null hypothesis that the posterior model probability was equal to the prior was neither clearly supported and nor clearly refuted by the Bayesian t-tests and the sensitivity analysis (see Fig. 4B), but showed annecdotal evidence for no bias. However, in the cases where a warning message was issued by the bridge sampler, we saw evidence for bias in the Bayes factor estimates, with average posterior model probabilities being too large (and larger than the prior probability; $BF10 > 3$).

Moreover, reliability diagrams (see Fig. 4C+D) again showed that for simulations without warning message it was unclear whether posterior model probabilities were the same as the prior probabilities. Specifically, for posterior model probabilities of around 50%, the prior model probability tended to be slightly low, but the 95% consistency bands did not provide clear evidence for bias. However, this was based on only very few simulations (see histgram in Fig. 4C), and more SBC simulations may be needed to draw firm conclusions. In

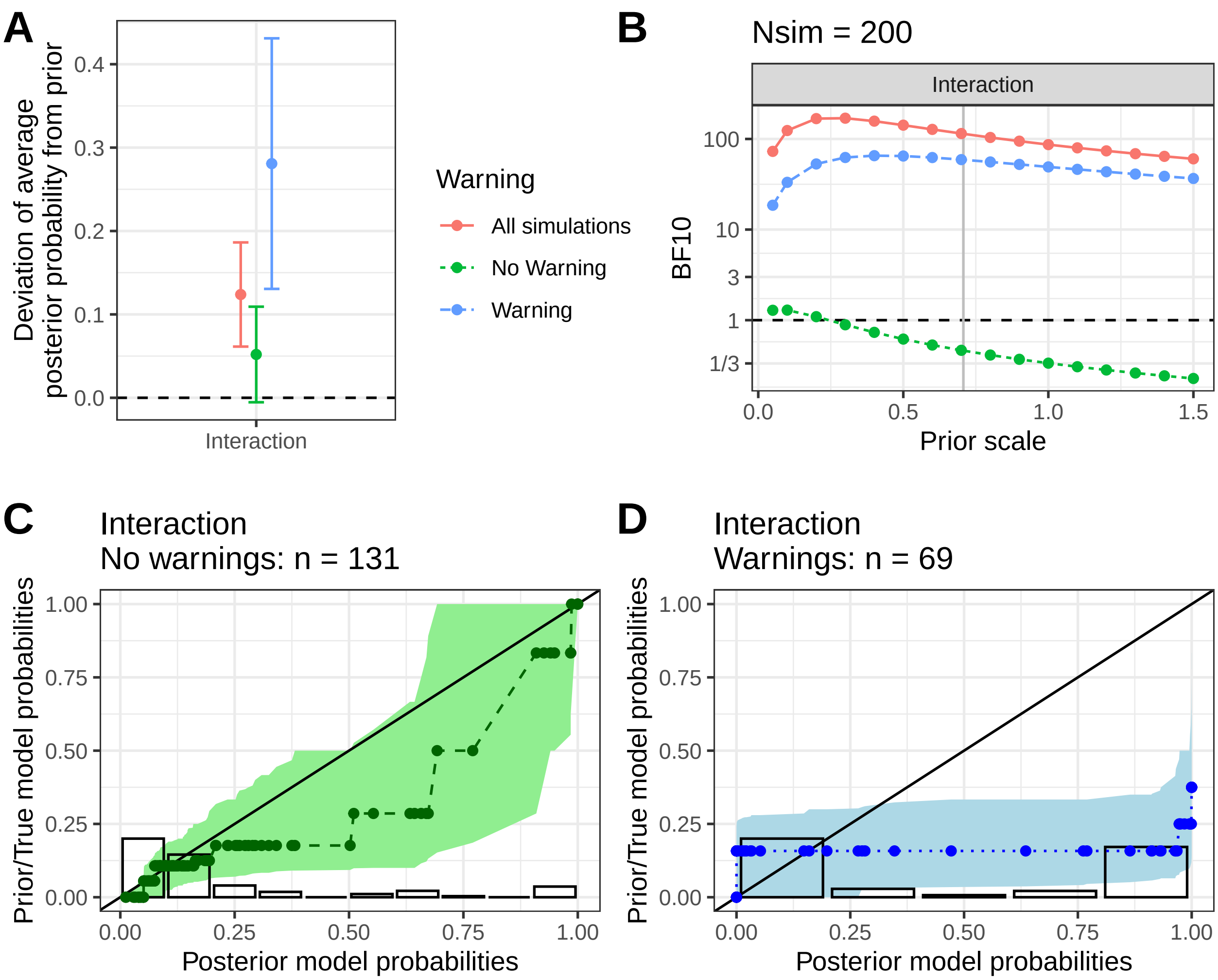

Figure 4. Repetition of the previous analyses of the interaction term (in design 3), but with 20,000 iterations of MCMC sampling instead of 10,000. A) The average deviation of posterior model probabilities from the priors (95% CI). All simulations and simulations without / with a warning message. Deviations from the horizontal broken line indicate bias. B) Bayesian t-tests of whether the posterior probability for H1 deviates from the prior, with sensitivity analyses for the prior scale. The vertical grey line indicates a default Bayes factor. C+D) Reliability diagrams: the prior model probability is plotted as a function of the estimated posterior model probabilities for simulations without (C) and with (D) a warning message. Error bands show 95% consistency bands. Results show that when no warning is issued, the results are unclear. When a warning message is issued, clear bias is visible.

simulations where a warning message was issued by the bridge sampler, the reliability diagrams suggested that posterior and prior model probabilities again did not match, indicating bias. Specifically, for high posterior probabilities (of H1) the prior probabilities were too low, again suggesting falsely high posterior model probabilities.

The results therefore suggest that doubling the number of MCMC iterations to $20,000$ was sufficient to slightly reduce the proportion of warning messages (from 37% to 34.5%), but does not suffice for substantially reducing the proportion of warning messages shown by the bridge sampling algorithm, and leads to similar results as with $10,000$ iterations. Potentially, even more MCMC iterations are needed to further reduce the proportion of warning messages issued by the bridge sampler.

Running 1000 SBC simulations. The previous results are limited in that we only performed 200 simulations of marginal SBC. Here, for better stability and power, we performed 1000 SBC simulations of the interaction term in the 2×2 repeated measures design using brms/bridgesampling.

Analysis of model convergence showed that across all model fits (1000 simulations $\times 2$ models (H0/H1) $= 2000$ fits), we observed 0 divergent transitions. R-hat exceeded the threshold of 1.01 for at least one population-level parameter in 2 cases, suggesting overall good model convergence. However, during the bridge sampling, there was the warning message “logml could not be estimated within maxiter, rerunning with adjusted starting value. Estimate might be more variable than usual” in 36.60% of simulations, again suggesting problems with the bridge sampling. We again looked at the results separately for simulations with versus without warning messages from the bridge sampler.

The results (see Fig. 5) showed that – for simulations without warning messages from the bridge sampling – the posterior model probabilities did not differ from the prior model samples. Indeed, the null hypothesis that the posterior model probability was equal to the prior was supported by Bayesian t-tests (see Fig. 5B) showing default Bayes factors of